

QUADRATIC EQUATIONS

Directions for question 1: Select the correct alternative from the given choices.

What are the factors of $6x^2 - 13x - 8$?

- ☐ a) $3x - 1, 2x + 8$
- ☐ b) $2x + 1, 3x - 8$
- ☐ c) $2x - 1, 3x + 8$
- ☐ d) $3x + 1, 2x - 8$

Solution:

$$6x^2 - 13x - 8$$

We have to split 6 into two parts, say a, b and 8 into two parts, say c, d such that $ad + bc = -13$. As -13 is odd all the two's in 6 and 8 have to go into only one term, either ab or bc.

$$\therefore 6x^2 - 13x - 8 = (2x + 1)(3x - 8).$$

Choice (B)

Directions for question 2: Type in your answer in the input box provided below the question.

Find the value of k, if the equation $2x^2 - 7x + k = 0$ has one root which is twice the reciprocal of the other.

Your Answer: Unattempted

Correct Answer: 4 

Solution:

$$2x^2 - 7x + k = 0 \dots(1)$$

Let the roots be α and $\frac{2}{\alpha}$.

$$\therefore \text{The product of the roots} = 2 = \frac{k}{2} \therefore k = 4$$

Ans : (4)

Directions for question 3: Select the correct alternative from the given choices.

What are the values of x that satisfy the inequality $3x^2 + 8x + 5 < 0$?

- ☐ a) $(-\frac{5}{3}, -1)$
- ☐ b) $(-1, \infty)$
- ☐ c) $(-1, -\infty)$
- ☐ d) $(-\frac{5}{3}, \infty)$

Solution:

$$3x^2 + 8x + 5 < 0$$

$$\Rightarrow (x + 1)(3x + 5) < 0$$

$$\Rightarrow (x + 1)(x + \frac{5}{3}) < 0$$

$$\therefore x \in \left(-\frac{5}{3}, -1\right)$$

Choice (A)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

For what value of p are the roots of the quadratic equation $px^2 - 18x + 81 = 0$ real and equal?

Solution:

$$\begin{aligned} px^2 - 18x + 81 &= 0 \quad \dots(1) \\ \text{As the roots of (1) are equal, } b^2 - 4ac &= 0, \\ \text{i. e. } \left(\frac{b}{2}\right)^2 - ac &= 0. \therefore \left(\frac{-18}{2}\right)^2 = 81p \Rightarrow p = 1. \\ &\text{Ans : (1)} \end{aligned}$$

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

A two-digit number is such that the product of the digits is 35. When 18 is added to the number, the digits interchange their places. Find the tens digit of the number.

Solution:

$$\begin{aligned} \text{Let the number be 'xy' } &= 10x + y. \\ \therefore xy &= 35 \quad \dots(1) \\ \text{And } 10x + y + 18 &= 10y + x \quad \dots(2) \\ (2) \Rightarrow 18 &= 9(y - x) \Rightarrow y - x = 2. \\ \text{Substituting } x + 2 &\text{ for } y \text{ in (1), we get } x(x + 2) = 35 \\ \therefore x &= 5 \quad (\because x \text{ can't be } -7). \text{ The number is 57.} \\ &\text{Ans : (5)} \end{aligned}$$

Directions for questions 6 to 11: Select the correct alternative from the given choices.

The two sides of a right-angled triangle containing the right angle are $3x + 4$ and $2x + 3$. If the area of the triangle is 20 sq. units, find the hypotenuse of the triangle.

- ☐ a) $\sqrt{89}$
- ☐ b) $\sqrt{116}$
- ☐ c) 15
- ☐ d) $\frac{\sqrt{3049}}{6}$

Solution:

The two legs of a right – angled triangle are $3x + 4$ and $2x + 3$. The area of the triangle is 20.

$$\therefore (3x + 4)(2x + 3) = 2(20)$$

$$\Rightarrow 6x^2 + 17x - 28 = 0$$

$$\Rightarrow (6x - 7)(x + 4) = 0$$

$$\Rightarrow x = \frac{7}{6} \text{ or } -4$$

If $x = -4$, the legs would have negative length.

This is not possible. $\therefore x = \frac{7}{6}$. The legs are $\frac{7}{2} + 4$ and $\frac{7}{3} + 3$, i.e. $\frac{15}{2}$ and $\frac{16}{3}$.

$\therefore$ If the hypotenuse is a ,

$$a^2 = \frac{225}{4} + \frac{256}{9} = \frac{2025 + 1024}{36} = \frac{3049}{36}$$

$$\therefore a = \frac{\sqrt{3049}}{6}$$

Choice (D)

Directions for questions 6 to 11: Select the correct alternative from the given choices.

Find the value of x if $\frac{2x}{3} + \frac{7}{3x} = 5$.

- ☐ a) 7
- ☐ b) $\frac{2}{3}$
- ☐ c) $\frac{1}{2}$
- ☐ d) More than one of the above

Solution:

$$\begin{aligned}\frac{2x}{3} + \frac{7}{3x} &= 5 \\ \Rightarrow 2x^2 - 15x + 7 &= 0 \\ \Rightarrow (2x - 1)(x - 7) &= 0 \\ \therefore x &= \frac{1}{2} \text{ or } 7.\end{aligned}$$

Choice (D)

Directions for questions 6 to 11: Select the correct alternative from the given choices.

What is the nature of the roots of the quadratic equation $-3x^2 + 4\sqrt{3}x - 15/4 = 0$?

- ☐ a) Irrational
- ☐ b) Real and equal
- ☐ c) Rational and unequal
- ☐ d) Imaginary

Solution:

$$-3x^2 + 4\sqrt{3}x - \frac{15}{4} = 0$$

$$\Rightarrow 3x^2 - 4\sqrt{3}x + \frac{15}{4} = 0$$

$$b^2 - 4ac = 48 - 3(15) = 3.$$

As the discriminant is positive, the roots are real and distinct. They are also irrational.
Choice (A)

Directions for questions 6 to 11: Select the correct alternative from the given choices.

The product of two alternate multiples of 4 is eight more than five times their sum. What are the two numbers?

- ☐ a) 4, 12
- ☐ b) 12, 20
- ☐ c) 8, 16
- ☐ d) 16, 24

Solution:

Let the alternate multiples of 4 be $4k$ and $4(k + 2)$

$$\therefore (4k) 4(k + 2) = 5(4k + 4k + 8) + 8$$

$$\Rightarrow 16k^2 + 32k = 40k + 48$$

$$\Rightarrow 16k^2 - 8k - 48 = 0$$

$$\Rightarrow 2k^2 - k - 6 = 0$$

$$\Rightarrow (2k + 3)(k - 2) = 0$$

$$\Rightarrow k = \frac{-3}{2} \text{ or } 2$$

As k is an integer, $k = 2$. The multiples are 8 and 16.

Choice (C)

Directions for questions 6 to 11: Select the correct alternative from the given choices.

The sum of an integer and three times its reciprocal is $301/10$. Find the integer.

- ☐ a) 60
- ☐ b) 10
- ☐ c) 30
- ☐ d) 50

Solution:

Let the integer be x

$$\therefore x + \frac{3}{x} = \frac{301}{10} = 30 + \frac{1}{10} = 30 + \frac{3}{30}$$

$\therefore x = 30$ (If x was not an integer, it could have been $\frac{1}{10}$). If we cannot immediately express the RHS as $30 + \frac{3}{30}$, we can proceed in the usual way. We would still get the answer.

Choice (C)

Directions for questions 6 to 11: Select the correct alternative from the given choices.

The sum of the squares of three consecutive odd positive integers is 251. Find the numbers.

- ☐ a) 11, 13, 15
- ☐ b) 9, 11, 13
- ☐ c) 7, 11, 13
- ☐ d) 7, 9, 11

Solution:

Let the consecutive odd integers be $x - 2$, x , $x + 2$.

$$\therefore (x - 2)^2 + x^2 + (x + 2)^2 = 251$$

$$\Rightarrow 3x^2 + 8 = 257 \Rightarrow x = \pm 9$$

The integers could be -11 , -9 , -7 or 7 , 9 , 11 .

But they have to be positive (given).

Choice (D)

Directions for question 12: Type in your answer in the input box provided below the question.

The number of diagonals that can be drawn in a polygon of x sides is given by $d = x(x - 3)/2$.

How many sides does a polygon have, if it has 104 diagonals?

Solution:

Let the number of sides of the polygon be x .

$$\therefore x(x - 3) = 2(104) = 2(8)(13) = 16(13)$$

$$\therefore x = 16 \text{ (as it cannot be } -13\text{).}$$

Ans : (16)

Directions for question 13: Select the correct alternative from the given choices.

In a particular month, the savings, in rupees of Suppandi are numerically equal to the positive square root of his expenditure in rupees. If his income for the month is ₹42, what is his expenditure?

- ☐ a) ₹64
- ☐ b) ₹25
- ☐ c) ₹49
- ☐ d) ₹36

Solution:

Let Suppandi's savings be x . His expenditure is x^2 .

$$\therefore x + x^2 = 42 = 6 + 6^2. \therefore x = 6.$$

Or proceeding in the usual way, we get $x^2 + x - 42 = 0$

$$\Rightarrow (x + 7)(x - 6) = 0. \text{ As } x > 0, x = 6.$$

Suppandi's expenditure is ₹36.

Choice (D)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

To do a work independently B requires six days more than A does. If they work together, they will take two days less than what A alone does. In how many days can B alone complete the work?

Solution:

Let A take x days to complete the work. B takes $x + 6$ days to do the same work.
Together they take $x - 2$ days to complete the work

$$\frac{1}{x} + \frac{1}{x+6} = \frac{1}{x-2}$$

$$\Rightarrow x^2 - 4x - 12 = 0$$

$$\Rightarrow (x+2)(x-6) = 0$$

$$\therefore x = 6.$$

B can do the work in 12 days.

Ans : (12)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

If the speed of a car is increased by 15 kmph, it takes one hour less to cover a distance of 300 km. Find the original speed of the car(in kmph)

Solution:

Let the original speed of the car be u km/hr

$$\therefore \frac{300}{u} - 1 = \frac{300}{u+15}$$

$$\Rightarrow 300(15) = u(u+15)$$

$$\Rightarrow 15(4)(5)(15) = u(u+15)$$

$$\Rightarrow (60)(75) = u(u+15) \therefore u = 60$$

(It cannot be -75)

Ans : (60)